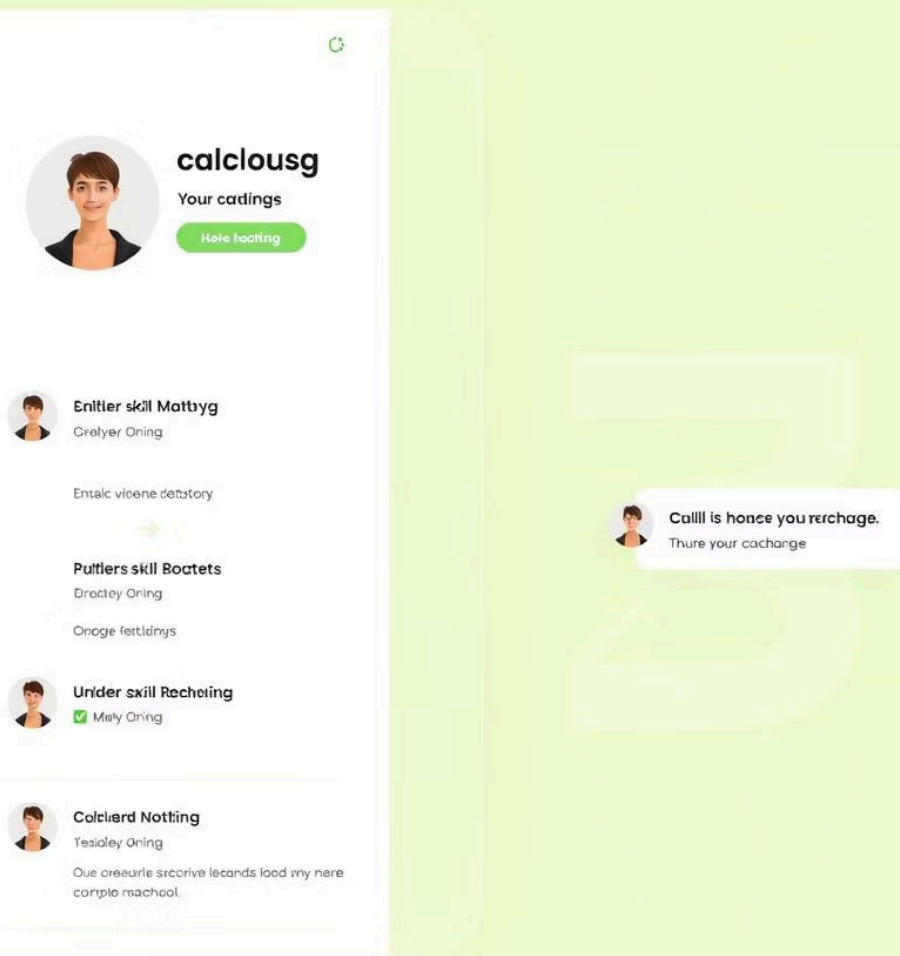




SkillSwap – Peer-to-Peer Learning Exchange

Welcome to SkillSwap, a platform revolutionizing peer learning. Teach your skills, learn new ones, and earn Skill Coins in a vibrant community.



What SkillSwap Does

Skill exchange platform

Trade knowledge without money using Skill Coins and gamification.

Smart matchmaking

Connect users based on mutual skills they want to teach and learn.

Built-in tools

Chat, schedule lessons, and track progress efficiently within the app.

The Problem We Solve

Access Barriers

Premium platforms are often unaffordable for many learners.

Knowledge Gaps

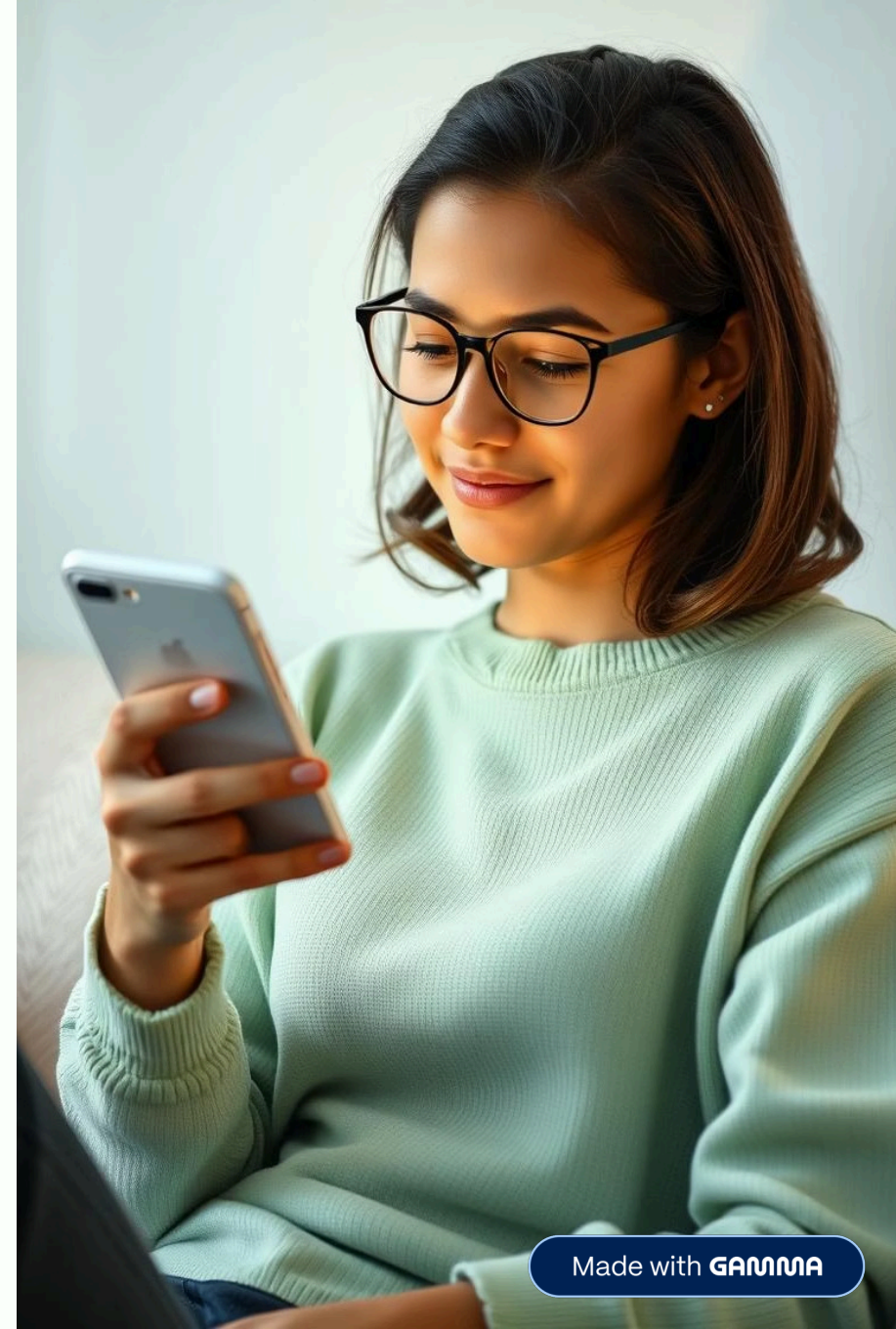
People nearby may hold valuable skills missing from online resources.

Incentive Issues

No easy way to trade and motivate knowledge sharing effectively.

Empowerment

Users want to learn while feeling empowered by teaching others.



Challenges We Faced

Matchmaking Algorithm

Balancing skill supply and demand for meaningful matches.

Chat & Scheduling

Integrating real-time communication with calendar synchronization.

Gamification

Creating a reward system that's engaging and secure from abuse.

Intuitive UX

Designing onboarding to be smooth for all skill levels.

Technology Stack



Frontend

React.js for web and Flutter for mobile apps



Backend

Node.js with Express, Firebase for real-time features



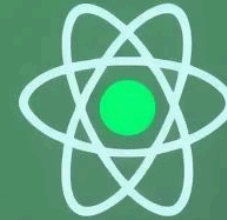
Database

Firestore and MongoDB for flexible data storage



AI (Optional)

Python + Flask + OpenAI API for skill matching



AI



How We Built SkillSwap

1

Brainstorm & Design

Identified needs and created wireframes on Figma

2

Backend Setup

Configured Firebase for authentication and real-time DB

3

Frontend Development

Built interfaces using React and Flutter

4

Feature Integration

Implemented chat, scheduling, and matchmaking logic

5

Gamification

Added Skill Coins and reward tracking system



Key Learnings

Community Power

Tech transforms peer learning into a vibrant, gamified system.

Real-Time Integration

Video chat and messaging work well within tight schedules.

Team Collaboration

Effective scoping and communication are crucial under pressure.

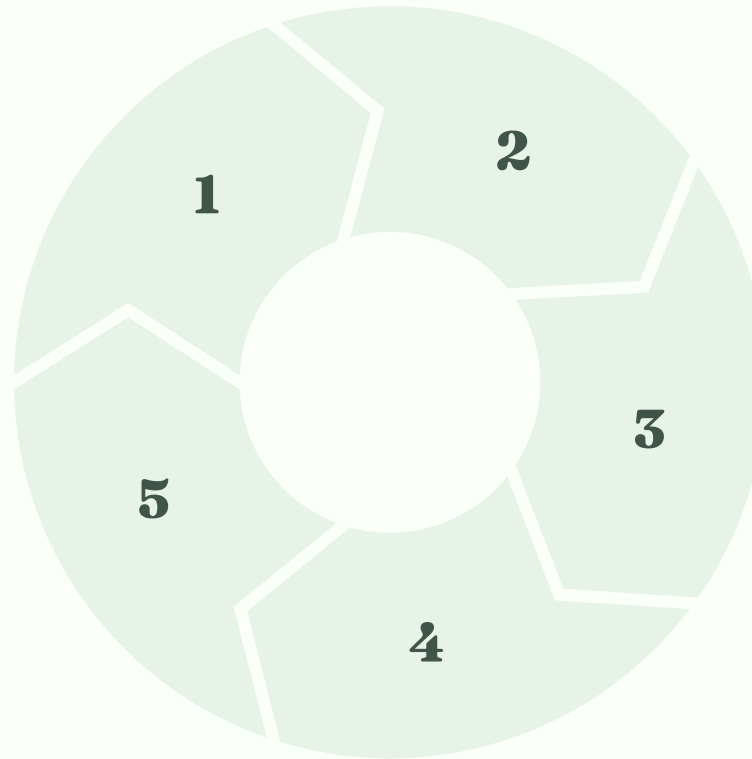
Next Steps for SkillSwap

AI Skill Assessment

Personalize paths using AI to measure proficiency

Group Sessions & Badges

Introduce community groups and mentorship rewards



Mobile App Launch

Enhance accessibility and engagement on smartphones

Multilingual Support

Expand inclusivity with regional languages

Partnerships

Collaborate with universities and NGOs for impact